His approach highlights the difference between a point-particle theory and string theory when such tears occur. The key distinction is that there are two types of string motion near the tear, but only one kind of point-particle motion. Namely, a string can travel adjacent to the tear, like a point particle does, but it can also encircle the tear as it moves forward, as illustrated in Figure 11.6. In essence, Witten's analysis reveals that strings which encircle the tear, something that cannot happen in a point-particle theory, shield the surrounding universe from the catastrophic effects that would otherwise be encountered. It's as if the world-sheet of the string—recall from Chapter 6 that this is a two-dimensional surface that a string sweeps out as it moves through space—provides a protective barrier that precisely cancels out the calamitous aspects of the geometrical degeneration of the spatial fabric.

You might well ask, What if such a tear should occur, and it just so happens that there are no strings in the vicinity to shield it? Moreover, you might also be concerned that at the instant in time that a tear occurs, a string—an infinitely thin loop—would provide as effective a barrier as shielding yourself from a cluster bomb by hiding behind a hula hoop. The resolution to both of these issues relies on a central feature of quantum mechanics that we discussed in Chapter 4. There we saw that in Feynman's formulation of quantum mechanics, an object, be it a particle or a string, travels from one location to another by "sniffing out" all possible trajectories. The resulting motion that is observed is a combination of all possibilities, with the relative contributions of each possible trajectory precisely determined by the mathematics of quantum mechanics. Should a tear in the fabric of space occur, then among the possible trajectories of travelling strings are those that encircle the tear-trajectories such as those in Figure 11.6. Even if no strings seem to be near the tear when it occurs, quantum mechanics takes account of physical effects from all possible string trajectories and among these are numerous (infinite, in fact) protective paths that encircle the tear. It is these contributions that Witten showed precisely to cancel out the cosmic calamity that the tear would otherwise create.

In January 1993, Witten and the three of us released our papers simultaneously to the electronic Internet archive through which